Overview

The data originates from the Mexican Government and contains roughly a million rows with twenty-one columns, each row of a distinct patient. This analysis seeks to further understand the leading causes of death within covid victims to better understand the prevention, protection, and treatment for said victims.

Data Preparation

The analysis begins with the integration of Python from Google BigQuery. Many rows were cleaned and organized to make working with it much more efficient. One of the problems encountered was the different index systems used by the Mexican government, leading to having to replace the data with more standard formats. Some rows used the values 97, 98, and 99 to refer to null values, which was changed to null for cleaner analysis. (Refer to *Fig. 1*)

Fig. 1

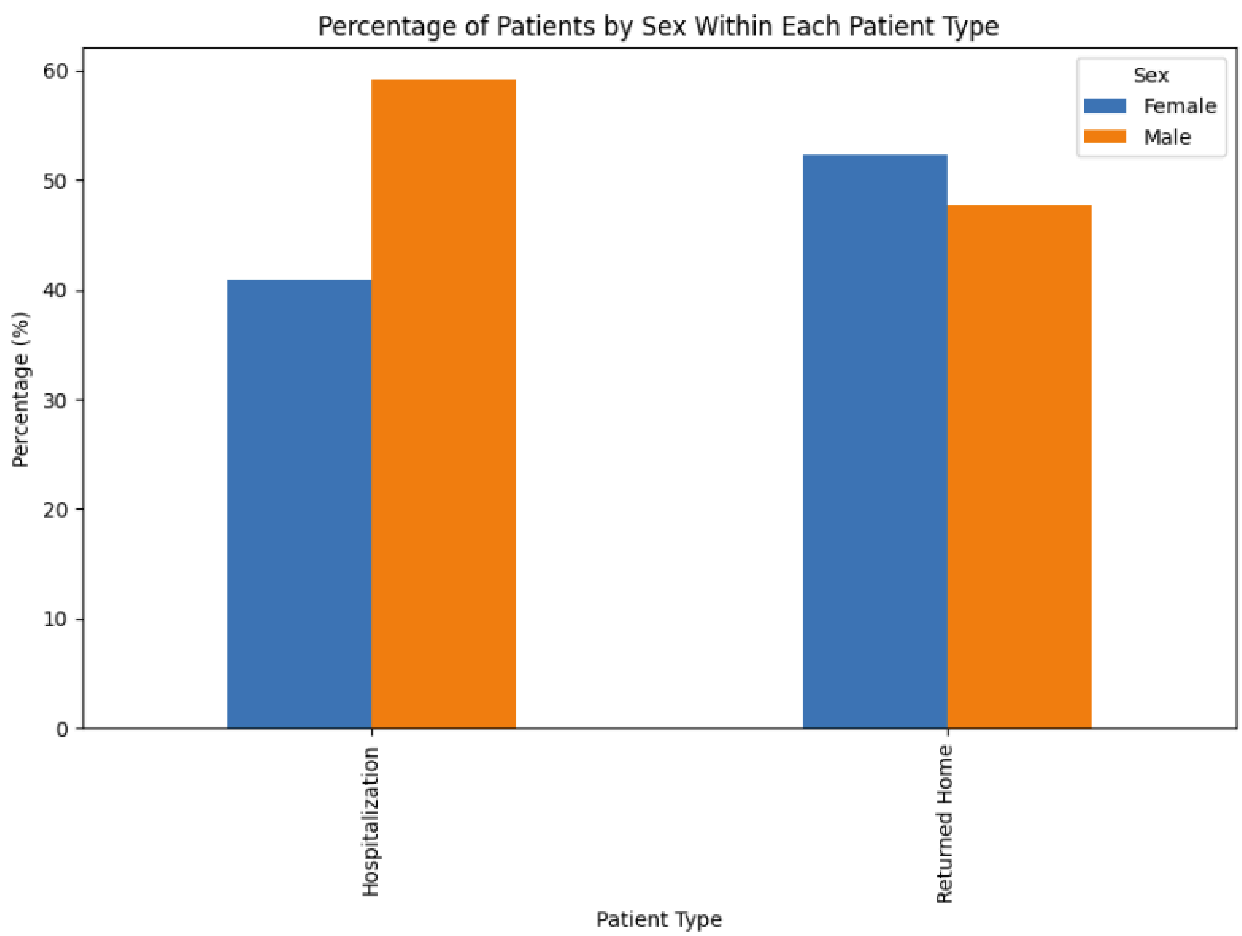
	USMER	MEDICAL_UNIT	SEX	PATIENT_TYPE	DATE_DIED	INTUBED	PNEUMONIA	AGE	PREGNANT	DIABETES	...	TOBACCO	CLASIFFICATION_FINAL	ICU	DATE_DIED_NEW	CLASSIFICATION_FINAL_NEW	HYPERTENSION	DIED	YEAR	MONTH	DAY
0	General Hospital	1	Female	Returned Home	03/05/2020	<NA>	1	65	0	0	...	0	3	<NA>	2020-05-03	Severe	1	1	2020	5	3
1	General Hospital	1	Male	Hospitalization	09/06/2020	1	0	55	<NA>	1	...	0	3	0	2020-06-09	Severe	0	1	2020	6	9
2	General Hospital	1	Female	Hospitalization	9999-99-99	0	1	40	0	0	...	0	3	0	NaT	Severe	0	0	<NA>	<NA>	<NA>
3	General Hospital	1	Female	Returned Home	9999-99-99	<NA>	0	64	0	0	...	0	3	<NA>	NaT	Severe	0	0	<NA>	<NA>	<NA>
4	General Hospital	1	Female	Returned Home	9999-99-99	<NA>	1	64	0	1	...	0	3	<NA>	NaT	Severe	1	0	<NA>	<NA>	<NA>
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	Primary Care	13	Male	Returned Home	9999-99-99	<NA>	0	28	<NA>	0	...	0	7	<NA>	NaT	Undiagnosed/Inconclusive	0	0	<NA>	<NA>	<NA>
1048571	Primary Care	13	Female	Hospitalization	9999-99-99	0	0	23	0	1	...	0	7	0	NaT	Undiagnosed/Inconclusive	0	0	<NA>	<NA>	<NA>
1048572	Primary Care	13	Male	Returned Home	9999-99-99	<NA>	0	47	<NA>	1	...	0	7	<NA>	NaT	Undiagnosed/Inconclusive	0	0	<NA>	<NA>	<NA>
1048573	General Hospital	13	Male	Returned Home	9999-99-99	<NA>	0	40	<NA>	0	...	0	7	<NA>	NaT	Undiagnosed/Inconclusive	0	0	<NA>	<NA>	<NA>
1048574	General Hospital	13	Male	Returned Home	9999-99-99	<NA>	0	28	<NA>	0	...	0	7	<NA>	NaT	Undiagnosed/Inconclusive	0	0	<NA>	<NA>	<NA>

1048575 rows × 28 columns

Exploratory Data Analysis

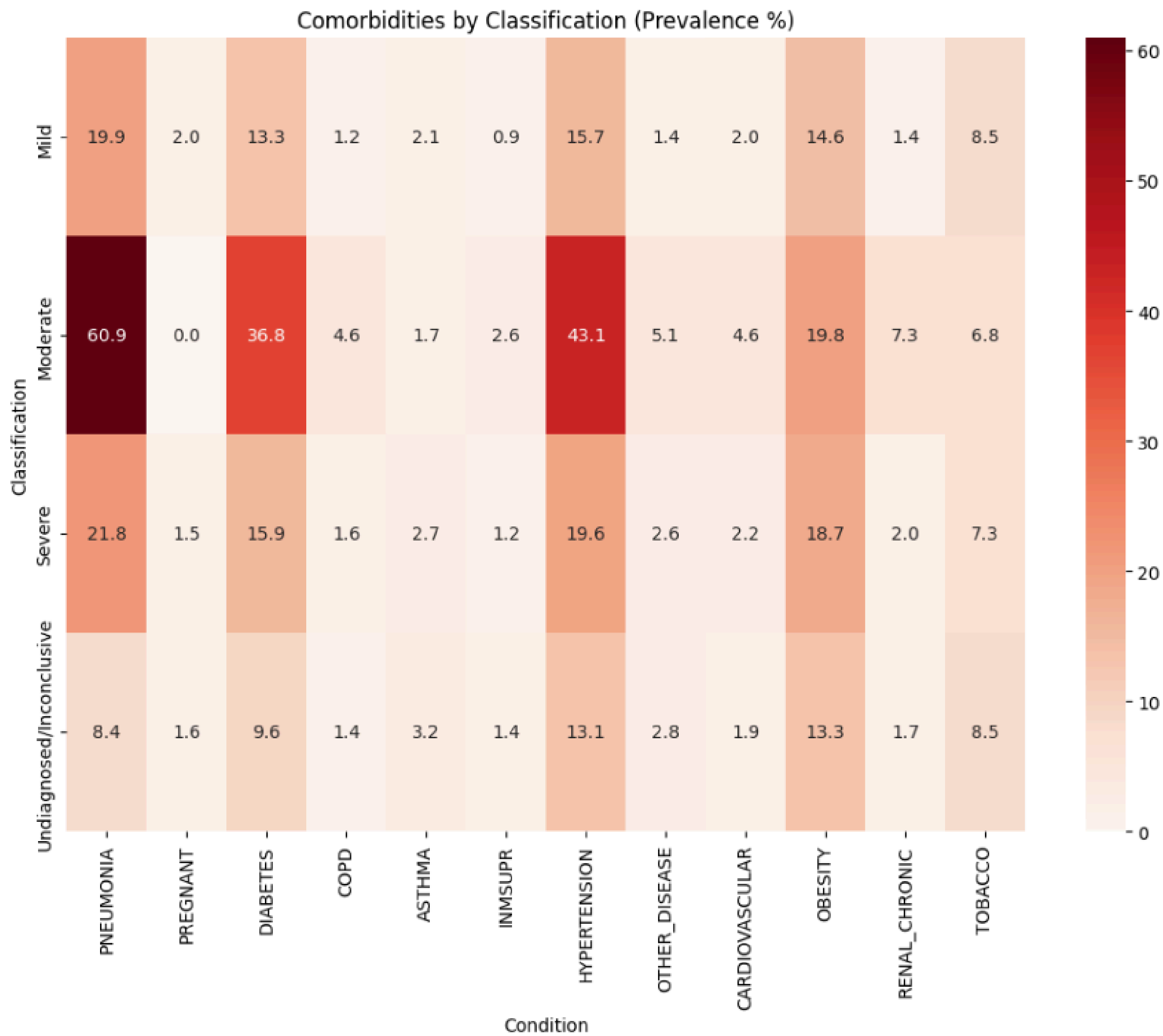
An EDA shows the percentage of males and females within the different categories of medical care facilities. A higher percentage of males appeared in the hospitalization category, which can be a potential reasoning for a future hypothesis relating to. (Refer to *Fig. 2*)

Fig. 2



Another EDA analysis showed the correlation between the different comorbidities by how severe it correlated with the different classifications of patients. The biggest finding is the high correlations between pneumonia and the classifications, at an average of 27.75 across all patient classifications. We will further explore this in a later machine learning model. (Refer to *Fig. 3*)

Fig. 3



Machine Learning Models

A logistic regression model was run to understand the correlations between death and the various comorbidities. We can see that pneumonia stands above the rest in terms of their correlation for death, with patients with pneumonia being 13.46 times more likely to die just from having pneumonia as a result from this model. Other comorbidities also contribute to death, but not as significant compared to pneumonia. (Refer to *Fig. 4*)

Fig. 4

```
model = sm.Logit(y, X)
result = model.fit(method="lbfgs", maxiter=200)

print(result.summary())
```

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                   DIED   No. Observations:              1026832
Model:                          Logit   Df Residuals:                  1026820
Method:                           MLE   Df Model:                           11
Date:                Tue, 16 Jun 2026   Pseudo R-squ.:                  0.3716
Time:                        18:12:33   Log-Likelihood:            -1.6865e+05
converged:                       True   LL-Null:                   -2.6840e+05
Covariance Type:            nonrobust   LLR p-value:                     0.000
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
const             -6.4742      0.018   -350.271      0.000      -6.510      -6.438
PNEUMONIA          2.6093      0.009    277.689      0.000       2.591       2.628
AGE                0.5566      0.003    174.633      0.000       0.550       0.563
DIABETES           0.4174      0.011     38.389      0.000       0.396       0.439
COPD              -0.1360      0.025     -5.550      0.000      -0.184      -0.088
ASTHMA            -0.3388      0.032    -10.588      0.000      -0.402      -0.276
INMSUPR            0.3577      0.029     12.422      0.000       0.301       0.414
HYPERTENSION       0.1689      0.011     15.455      0.000       0.147       0.190
CARDIOVASCULAR    -0.1423      0.023     -6.196      0.000      -0.187      -0.097
OBESITY            0.2282      0.011     19.922      0.000       0.206       0.251
RENAL_CHRONIC      0.6735      0.022     30.985      0.000       0.631       0.716
TOBACCO           -0.0818      0.017     -4.849      0.000      -0.115      -0.049
==================================================================================
```

A z-test was also run due to the large size of the dataset, used to target the specific variable of pneumonia and their correlation to death. A p-value of 0.0 and a z-statistic of 479.18 is statistically significant in proving the direct correlation between pneumonia and death. With further analysis, we concluded that the death rate for patients with pneumonia is 38.51% while for patients without pneumonia is 2.5% making patients 15.42 times more likely to die from contracting pneumonia while having covid, even more of a significant risk than we previously discovered. (Refer to *Fig. 5*)

Fig. 5

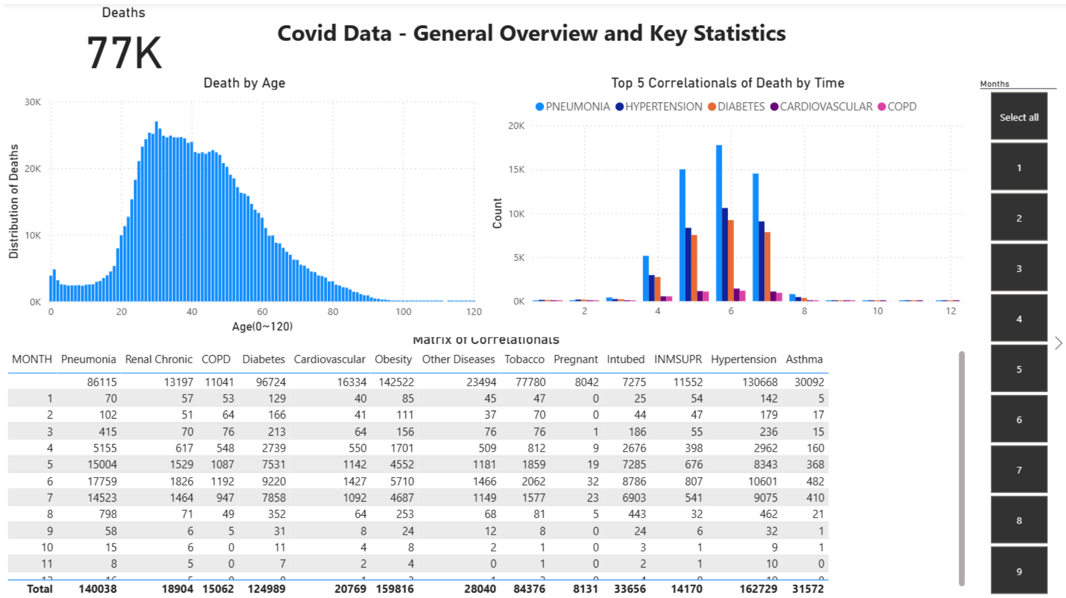
```python
#Preparing for Z-test

from statsmodels.stats.proportion import proportions_ztest

# Clean data
df_test = df_clean[["DIED", "PNEUMONIA"]].dropna()

pneumonia = df_test[df_test["PNEUMONIA"] == 1]["DIED"]
no_pneumonia = df_test[df_test["PNEUMONIA"] == 0]["DIED"]

count = np.array([
    pneumonia.sum(),
    no_pneumonia.sum()
])

nobs = np.array([
    len(pneumonia),
    len(no_pneumonia)
])

#Z-Test
z_stat, p_value = proportions_ztest(count, nobs)

print("Z-statistic:", z_stat)
print("P-value:", p_value)
```

```
Z-statistic: 479.1776178264977
P-value: 0.0
```

Data Visualization

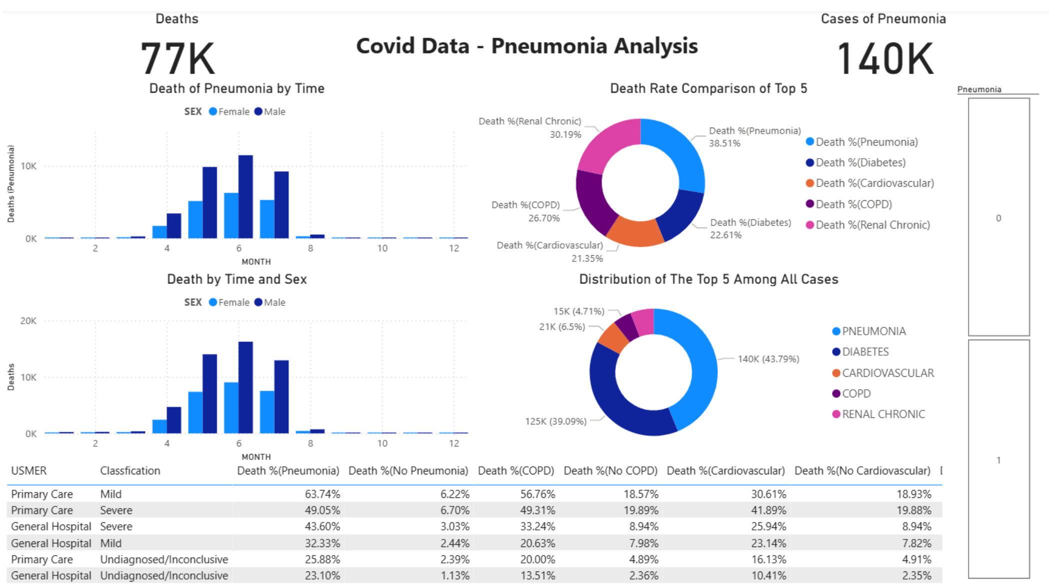
A dashboard was produced to inform about the general information that circled around the deaths Covid-19 produced. With significant outbreaks of death and comorbidities occurring between April of 2020 to August of 2020. (Refer to *Fig. 6*)

Fig. 6



A second dashboard was produced focusing more closely on the prospect of pneumonia. With death rates for patients sky rocketing under the presence of pneumonia, affecting males at a significantly higher rate. (Refer to *Fig. 7*)

Fig. 7



Conclusion

Pneumonia is a significant factor in the prevention, protection, and treatment of Covid-19 victims. Being present among the most important matrices and contributing to the overall danger to wellness among patients. Males seem to be disproportionately affected by Covid-19. This analysis can lead to better preventions and treatments of the disease now that we understand the most dangerous factor that contributed to the most significant risk.